

PROJECT LOG & TECHNICAL ARCHIVE

Vintage Machinery Restoration • Motor Housing & Electrical Base

Project Target:	Vintage Cast-Aluminum Machine Motor & Control Base
Current Phase:	Housing Post-Paint Curing & Electrical System Verification
Archive Date:	June 2, 2026

1. Overview of Restored Components

This archive tracks the multi-phase technical restoration of a mid-century cast-aluminum machine motor assembly and its integrated control base. The project balances mechanical remediation, authentic cosmetic refinishing, and safety-minded electrical modernizations.

- Motor Housings:** Two cast-aluminum dome casings housing the stator and rotor assemblies. These parts require durable high-temperature coatings and meticulous bearing-bore tolerances.
- Main Control Base Casting:** The cast-aluminum structural floor plate containing the factory control hardware, speed management circuits, and internal switching links.

2. Paint & Cosmetic Refinishing Milestones

Surface Preparation & Etching

The bare aluminum components were meticulously degreased using 99.9% Isopropyl Alcohol (IPA) to eliminate all structural grease and legacy contamination. A professional zinc phosphate self-etching primer was laid down in ultra-thin, continuous coats. This created a highly stable chemical anchor, visible as a completely flat, non-reflective matte olive-drab/grey layer, protecting against future oxidation and flash lifting.

Aerosol Texture Deposition (VHT Wrinkle Plus)

Following a tight 15-minute solvent flash window, the VHT Wrinkle paint was applied in a precise three-coat crosshatch matrix to provide structural depth:

- Coat 1:** Solid vertical passes ensuring complete wet edge coverage.
- Coat 2:** Full horizontal cross-directional passes to interlock the paint binders.
- Coat 3:** A thick final uniform wet coat providing the chemical mass needed to texturize.

Thermal activation was initiated after a 3-minute level window using a bare-nozzle heat gun operating at 350°F to accelerate chemical cross-linking. This successfully pulled out a tight, pristine, highly uniform heavy grain structure mimicking the original factory finish. Masking layers were pulled cleanly while the film was still warm, saving internal mechanical dimensions.

Post-Cure Observation & Conditioning Notes

Current Phase: The components are currently undergoing a mandatory passive internal cure, isolated on their assembly board to protect the soft core matrix from compression marks or finger indentations.

Sooty Appearance: The current flat, dry, powder-like appearance resembling carbon soot is a normal transient phase caused by raw light refraction across fresh, highly vertical micro-peaks. Over a 48-hour window, deep chemical shrinkage will settle the ridges into a rounded leather grain with a native semi-gloss satin sheen.

Remediation Protocols (Friday Window): If localized heavy accumulation areas fail to settle, micro-peaks can be dry-scuffed using 400-600 grit wet/dry paper or an ultra-fine gray Scotch-Brite pad. Trapped micro-overspray dust can be cleared with a medium nylon bristle brush. Total cosmetic deep-black equalization can be locked in by applying a microscopic film of mineral oil or WD-40 wiped off completely via microfiber cloth.

3. Electrical Infrastructure & Circuit Logic

The main base casting houses an legacy non-grounded, 110V AC electrical system that relies entirely on structural component isolation for operator protection. The incoming electrical cord is a legacy 2-wire arrangement (Hot and Neutral) without an integrated protective earth conductor.

Component Matrix:

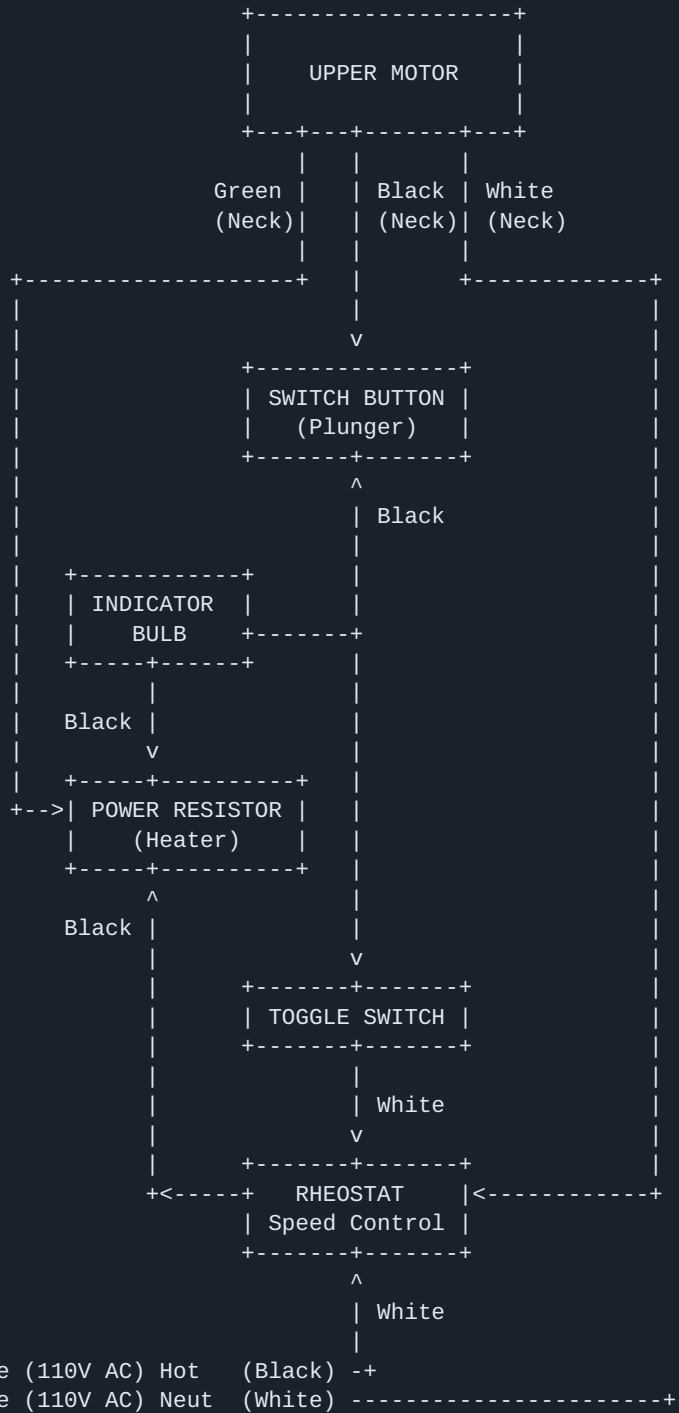
- **The Rheostat:** A ceramic-insulated toroidal power resistor wound with heavy resistance wire. It manages motor speed variables by modifying voltage returns via a sweeping wiper arm.
- **The Plunger Switch Button:** A top-mounted momentary mechanical switch responsible for inline distribution of Hot AC line power directly to the upper motor casing.
- **The Power Resistor ("Heater"):** A massive, ceramic-bodied inline drop resistor designed to step down full line voltage into a low-current stream for the status indicator light.
- **The Status Bulb & Toggle Switch:** A chassis-mounted incandescent indicator lamp sitting behind a faceted red jewel lens, coupled to a legacy side-mounted toggle safety override or mode switch.

Critical Rebuild Hazard Identification

The Neck Wiring Illusion: The internal wire loom passing upward through the casting neck to the motor contains three wires color-coded Green, Black, and White. Testing and tracing reveal that the **internal Green wire is NOT a safety ground**. It is a live, current-carrying branch link connecting the motor assembly down to a dedicated terminal block on the variable rheostat stack. During rewire, this must remain isolated as a hot circuit element.

4. Master System Electrical Schematic

The schematic below documents the exact functional wiring pathways currently observed in the base layout, cross-referenced from the physical traces and user line mapping.



5. Upcoming Modern Safety Integration (3-Prong Upgrades)

To mitigate shock hazards and ensure compliance with modern electrical standards, a primary recommendation for the reassembly phase is the conversion of the main base input into a true 3-wire grounded topology:

1. **Cordage Specification:** Procure a high-grade 3-conductor cord (14AWG or 16AWG) complete with a molded NEMA 5-15P grounding plug.
2. **Current-Carrying Paths:** Terminate the incoming Black (Hot) and White (Neutral) leads exactly as designated in the historical blueprint.
3. **Chassis Bond Establishment:** The incoming Green ground wire from the cord must be permanently bonded directly to the aluminum base housing. This will be accomplished by selecting a structural screw location (such as the internal cable strap retention bosses), stripping the localized contact area down to clean, bright aluminum metal, and securing the lead using a heavy-duty crimped brass ring terminal. This provides an immediate low-impedance faults path directly back to the household breaker box.